# W10: Workshop Exercise: Artificial General Intelligence & Risk

*TEXT FROM: STEVE CLARKE (2005)*

## *The issue.*

The prospect of far reaching developments in artificial intelligence, nanotechnology and genetic engineering have prompted the expression of fears about possible catastrophic scenarios. For example, Drexler (1986) argues that developments in nanotechnology may lead to the creation of self-assemblers that will be able to convert everything in the universe into themselves. Eventually the universe will be composed only of self-assemblers. This is known as the "grey goo" problem (Drexler 1986: 172).

[ . . . ]

It is extremely hard to say exactly how likely it is that such dystopic possibilities will occur, even if it seems safe to say that they are most unlikely. It is also extremely hard to know how to assess the costs of such possibilities, even if it seems safe to say that they are astronomically high. How are we to factor these catastrophic possibilities into policy-making? Here

## *The "Precautionary Principle.*

A third approach is to ignore other considerations, in the face of such possibilities and, taking the attitude "better safe than sorry", institute a moratorium, or some other form of restriction, on research that might lead to such dystopic scenarios.

[ . . . ]

What different versions of the precautionary principle have in common is that they stress that we should be willing to act to avoid bringing about a possible set of undesirable circumstances, even when we lack firm evidence to suggest that such an outcome is even possible.

One well-known and influential formulation of the precautionary principle is the Wingspread Statement of 1998:

> Where an activity raises threats of harm to the environment or human health, precautionary measures should be taken even if some cause and effect relationships are not fully established scientifically (Wingspread 1998).

[ . . . ]

## *An Argument Against the Precautionary Principle.*

Manson (2002: 270–274) and Sunstein (2002: 102–104) both argue that stronger versions of the precautionary principle entail a paradox and lead to contradictory policy recommendations. To appreciate the paradox, consider again the grey goo problem. It is possible, as Drexler (1986: 172–173) argues, that our efforts to conduct research in nanotechnology will cause the grey goo problem. However, it is also possible that our failure to conduct research in nanotechnology will lead indirectly to the grey goo problem, or some other catastrophic scenario. For example, it seems possible that research in nanotechnology that we did not conduct might have led to us finding a way to prevent the grey goo problem and it seems possible that on another planet in the universe a civilization flourishes that conducts research into nanotechnology that leads to the creation of self-assemblers that begin to convert everything into themselves. If we fail to conduct research that leads us to find a way to prevent the grey goo problem and if the alien civilization in our example does conduct research that leads to the creation of self-assemblers that begin to convert everything into themselves, then our failure to conduct research in nanotechnology will have contributed to causing a catastrophe, because we will be unable to prevent the grey goo problem. So the precautionary principle leads us to conclude both that we should conduct research into nanotechnology and that we should not conduct research into nanotechnology.

Text taken from:
Clarke, Steve (2005). "Future Technologies, Dystopic Futures and the Precautionary Principle". *Ethics and Information Technology* 7: 121-126.

***(1.)What is the "Precautionary Principle? (PP)***

***(1..a) What kind of situations does PP pertain to?***

***(1.b.) What does PP tell us to do in such situations?***

***(2.) Try Summarizing the argument against PP.***

P1. If PP is true, then___

***(3.) Is there any rationale/support that can be offered for one or more key premise?***

***(4.) Does the anti-PP argument seem strong? Can you think of a way a defender of PP might respond?***